CropState-Vision: A Reliability-Oriented Evaluation of Six-Stage Rice Growth Classification and Confidence-Aware Stage-Gated Retrieval

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***Abstract*—CropState-Vision studies six morphology-derived, BBCH-inspired rice pseudo-stages and uncertainty-aware evidence retrieval. A three-seed fixed-split ResNet-18 evaluation on 2,900 RiceSEG-derived crops reaches 0.800 mean accuracy against weak labels, not expert-confirmed phenology. We correct the retrieval protocol to use English BM25 queries for 173 English-language IRRI chunks, and evaluate 420 metadata-labeled scenarios from 199 crops belonging to 63 original photographs. Adaptive gating yields nDCG@5 of 0.545 versus 0.343 for query expansion and metadata-defined SIRR@5 of 0.127 versus 0.301. Hard filtering has SIRR@5 of 0.011 but does not return five documents in 10% of scenarios. Validation selects fixed-soft $\beta = 0.35$, which reaches nDCG@5 0.612 / SIRR@5 0.101, and a top-1-belief variant reaches 0.617 / 0.098; both outperform combined-confidence adaptive gating on these metadata metrics. Paired uncertainty estimates resample the 63 original parent photographs, with Holm-adjusted tests. Results establish an internally consistent evaluation of stage-conditioned retrieval; neither expert-judged BBCH accuracy nor agronomic usefulness or safety is demonstrated.**

***Index Terms*—rice growth stage, BBCH, image classification, ordinal loss, calibration, selective prediction, stage-aware retrieval, data leakage**

I. Introduction

Rice management recommendations depend on growth stage: irrigation, nutrient scheduling, disease surveillance and harvest preparation must follow plant development rather than the calendar. Crop-stage recognition from field images is challenging near phenological transitions and under changes in lighting, viewpoint and plant density. Overconfident mistakes can propagate into retrieval if a top-1 stage prediction is treated as certain.

CropState-Vision retains a six-stage probability distribution and uses it to adjust metadata-driven evidence reranking. The aim is a reproducible, reliability-oriented *evaluation*, not a novel vision backbone, a new retrieval model, or an autonomous agronomic advisor.

We address three linked questions: (RQ1) What can be concluded about six-stage visual classification and selective prediction on a fixed, near-duplicate-audited split? (RQ2) What ranking versus stage-incompatibility trade-offs arise for hard, fixed-soft and confidence-adaptive retrieval? (RQ3) Which evidence limitations constrain these findings? Our contributions are:

1) A group-aware aHash duplicate audit and three-seed ResNet-18 comparison, with ordinal, class-imbalance and calibration reporting on *morphology-derived* reference labels.
2) An English-query, metadata-defined retrieval benchmark of 420 topic scenarios, using 63 original-parent-photo clusters for paired bootstrap and sign-flip inference.
3) Validation-only comparison of tuned fixed-soft weights, full-belief, top-1, entropy and margin variants, including unfavorable results for the original adaptive formula and underfilled hard-filter outputs.
4) Explicit boundaries: weak phenology labels, small rare-stage classes, a single dataset split, and relevance scores sharing stage metadata with the reranker.

II. Related Work

The BBCH scale gives standardized numerical descriptions of plant development [1]. We group fine-grained rice codes into six macro-stages as a classification convenience, not a BBCH replacement. Machine learning has been applied to rice development from field imagery [2]; RiceRes2Net detects panicles and recognizes three reproductive sub-stages [3], and a lighter YOLO version ships as a smartphone app [4]. Both target seedling- or panicle-level sub-stages, whereas CropState-Vision classifies whole images across the vegetative-to-ripening cycle. A random split can overstate field generalization if acquisition cues leak across train/test [5].

ResNet is the pretrained backbone used here [6]; EfficientNet and Vision Transformers exist but were not evaluated, since the focus is training objective, not architecture search. Temperature scaling checks calibration [7], and selective classification lets a model abstain on uncertain cases [8]. Predictive uncertainty is known to degrade under distribution shift [9], and the aleatoric/epistemic distinction [10] motivates treating confidence as a routing signal rather than a correctness guarantee, the stance taken throughout this paper. Focal loss [11] is a usual response to class imbalance. The six stages are ordered, not nominal, motivating a distance-aware objective

TABLE I
RICE MACRO-STAGES USED IN THE STUDY.

Group	BBCH	Label
Germination/leaf development	00–19	Establishment
Tillering	20–29	Tillering
Stem elongation/booting	30–49	Stem/booting
Heading and flowering	50–69	Reproductive
Grain development	70–79	Grain filling
Ripening	80–89	Ripening

and MASD alongside accuracy [12]. Statistically, we use bootstrap intervals, Holm correction, and McNemar's test [13].

Retrieval-augmented generation grounds language-model output in external evidence [14]: BM25 for lexical matching [15], dense retrieval for semantic similarity [16], fused by Reciprocal Rank Fusion [17]. CropState-Vision follows this recipe with a sentence-transformer retriever [18], adding confidence-aware stage re-ranking. Against the closest rice-stage studies – [2], [3], [4], which cover only 3 seedling or panicle-level sub-stages and report neither calibration nor significance testing – CropState-Vision covers all 6 macro-stages and reports ECE/Brier, MASD, and bootstrap/McNemar testing.

III. TASK FORMULATION

A. Six Rice Macro-Stages and Stage Belief

Let $\Phi = \langle \phi_1, \ldots, \phi_6 \rangle$ be the ordered stage set in Table I. For image I, model f_θ produces logits $o = f_\theta(I)$ and the stage distribution $p(\phi_i \mid I) = \mathrm{softmax}(o/T)_i$, with $T > 0$ from validation logits when temperature scaling applies. The top-1 prediction is $\hat{\phi} = \arg\max_i p(\phi_i \mid I)$, but we retain the full belief $B = p(\cdot \mid I)$, summarized by normalized entropy $\Gamma = 1 - H(B)/\log 6$ and top-two margin $M = B_{(1)} - B_{(2)}$, combined as

$$p(\phi_i \mid I) = \mathrm{softmax}(o/T)_i, \quad \kappa = \eta\Gamma + (1-\eta)M, \qquad (1)$$

with $0 \le \eta \le 1$. Here κ is a routing signal, not proof the predicted stage is correct.

B. Confidence-Aware Stage-Gated Retrieval

Each chunk x carries stage-compatibility $C(x, \phi_i) \in [0, 1]$ (1 = direct, intermediate = adjacent, 0 = incompatible). BM25 and dense retrieval give candidate lists fused by RRF and normalized to $S_{\mathrm{base}}(x)$. Expected compatibility, the confidence-scaled stage weight, and the final score are

$$\begin{aligned} S_{\mathrm{stage}}(x, B) &= \sum_{i=1}^{6} B(\phi_i)\, C(x, \phi_i), \\ \beta &= \beta_{\min} + (\beta_{\max} - \beta_{\min})\, \kappa, \\ S(x) &= (1-\beta)\, S_{\mathrm{base}}(x) + \beta\, S_{\mathrm{stage}}(x, B). \end{aligned} \qquad (2)$$

β is deliberately linear in κ rather than a sigmoid or learned function: with only three seeds and 420 evaluation scenarios there is not enough held-out data to fit a nonlinear mapping without overfitting the operating point, and a linear rule keeps the confidence-to-weight relationship auditable. Unlike top-1 hard filtering, $S(x)$ does not exclude a chunk solely for being incompatible with the most probable stage. Metadata-defined stage incompatibility (not real-world agronomic safety) is measured by

$$\mathrm{SIRR@}k = \tfrac{1}{k} \sum_{x \in \mathrm{TopK}} \mathbb{I}[C(x, \phi^*) = 0], \qquad (3)$$

where ϕ^* is the morphology-derived reference pseudo-stage; the denominator k is unchanged if fewer than k results are returned.

TABLE II
NEAR-DUPLICATE AUDIT OVER THE 2,900-IMAGE BENCHMARK.

Hamming thresh.	Near-dup. edges	Multi-img. clusters	Straddling clusters	Cross-split pairs
10	661	122	0	0
12	2,173	131	32	197
14	6,712	68	21	999

IV. MATERIALS AND METHODS

A. Benchmark Construction and Near-Duplicate Audit

RiceSEG supplies segmentation masks rather than expert-assigned BBCH stages [19]. We use 2,900 eligible RGB/mask crops from the 3,078 annotated source images. Six BBCH-inspired pseudo-stage labels are derived from segmentation statistics rather than direct plant observations: a panicle fraction cutoff of 0.02 separates panicle-bearing crops; senescence fractions of 0.05 and 0.20 distinguish reproductive, grain-filling and ripening categories. For crops without emerged panicles, canopy-cover fractions of 0.31 and 0.51 distinguish establishment, tillering and stem/booting. These are dataset-specific heuristics, not expert-verified stages.

The fixed split contains 2,400 training, 237 validation and 263 test crops, shared by training seeds 42, 7 and 123. Test support is uneven: establishment 54, tillering 88, stem/booting 64, reproductive 24, grain filling 10, ripening 23. The aHash (grayscale 8×8) Hamming audit reports 661 near-duplicate edges at distance ≤ 10 and zero cross-split matches at that threshold (Table II). The split was grouped by detected near-duplicates; zero audited crossings therefore cannot establish independence across fields, seasons or devices.

B. Vision Models and Training Objectives

The model is an ImageNet-pretrained ResNet-18 (224-pixel input, batch 32, at most 30 epochs, AdamW; head/backbone learning rates $3 \times 10^{-4}/3 \times 10^{-5}$; frozen backbone for the first three epochs). All objectives use one fixed split and three training seeds; the checked-in configuration uses class-weighted losses but disables balanced sampling. We compare cross-entropy, an ordinal-aware loss on indices $0, \ldots, 5$,

$$L_{\mathrm{ord}} = L_{\mathrm{CE}} + 0.5\big(\mathbb{E}[y] - y^*\big)^2, \qquad (4)$$

and a weighted focal variant ($\gamma = 2$). Its implementation applies $\exp(-L_{\mathrm{CE,weighted}})$ in the modulation factor and is

TABLE III
RETRIEVAL METHODS COMPARED.

ID	Description
B0	Ungated BM25+dense retrieval, normalized RRF.
B1	Query expansion: append top-1 predicted stage.
B2	Hard filtering: remove zero-compatibility chunks.
B3	Soft re-ranking, untuned fixed $\beta = 0.20$.
P	Combined-confidence, full-belief adaptive re-ranking.

TABLE IV
RESNET-18 RESULTS ON THE 2,400/237/263 SPLIT (MEAN $\pm$ SD OVER 3 SEEDS).

Model	Acc.	F1	MASD↓	ECE↓	Rep.rec.
Majority	0.335	0.084	1.10	n/a	0.000
CE	$0.800\pm.007$	$0.794\pm.019$	$0.238\pm.013$	$0.062\pm.014$	$0.903\pm.052$
Ordinal	$0.801\pm.013$	$0.796\pm.018$	$0.233\pm.013$	$0.064\pm.002$	$0.917\pm.000$
Focal	$0.767\pm.016$	$0.757\pm.014$	$0.271\pm.010$	$0.179\pm.027$	$0.875\pm.059$

not the standard unweighted focal expression [11]. Reported vision metrics are historical fixed-split results; the corrected retrieval experiment does not retrain these vision checkpoints.

C. Retrieval Configuration and Baselines

The controlled test uses 173 non-restricted English-language IRRI knowledge chunks [20]. BM25 queries are now in *English*, matching the corpus, while multilingual MiniLM-L12-v2 dense embeddings use a verified FP32 ONNX export (128-token cap, masked mean pooling, L2 normalization). We fuse lexical/dense lists with RRF ($k_0 = 60$, depth 50), use topic-exact candidates and set source-authority weight to zero. The earlier Vietnamese lexical queries gave zero BM25 token overlap and their numeric results are superseded.

A scenario is eligible when a stage/topic pairing has at least three relevant and three incompatible chunks, with metadata relevance defined at compatibility ≥ 0.5. Thus 420 scenarios arise from 199 of 263 test crops, nested in 63 original photographs. Stages and judgments use metadata also consumed by the reranker; they measure internal consistency, *not* independent advice usefulness. Combined adaptive confidence uses entropy concentration (weight 0.6) and top-two margin (0.4), with $\beta_{\min} = 0.05$ and $\beta_{\max} = 0.35$; fixed B3 has untuned $\beta = 0.20$. The reference-stage oracle uses the weak reference pseudo-stage, not an ideal relevance ranking. Table III defines the alternatives.

D. Metrics and Statistical Analysis

Vision metrics include accuracy, macro-F1, ECE, Brier score, Mean Absolute Stage Distance (MASD) and per-class recall against weak pseudo-stage labels. Selective risk is one minus accuracy over retained weak-label test crops. We compute full risk–coverage curves at all 263 sorted top-k prefixes. The exact full-coverage AURC is the mean prefix risk $\frac{1}{263}\sum_{k=1}^{263} R(k)$, rather than a trapezoid over sparse manually chosen thresholds. Retrieval metrics are P@5, R@5, nDCG@5 [21] and metadata-SIRR@5. Unlike crop-wise resampling, the paired bootstrap and sign-flip tests cluster all topics/crops belonging to the same *original parent photograph* (63 groups), with 5,000 resamples and Holm adjustment over 12 prespecified method–metric comparisons [22]. Incomplete Top-5 returns are measured separately because SIRR@5 divides by five even when fewer chunks are returned.

V. RESULTS

A. Stage Classification (RQ1)

All three objectives clear the majority reference by a wide margin (Table IV); differences fall within seed-to-seed noise. Ordinal edges ahead slightly on average, CE keeps the lowest ECE, but one run can flip the ranking (seed 42 alone: ordinal 0.814, CE 0.795).

The ordinal objective yields identical reproductive recall of 0.917 across seeds on the *same 24 test crops*, compared with CE standard deviation 0.052. This is conditional training variability, not evidence of general stability across independent fields or data splits. Grain filling has only ten test crops; focal yields lower accuracy and worse calibration *under its implemented weighted formulation*. Seed-42 CE versus ordinal comparisons show no statistically resolved accuracy difference ($p = 0.40$) or MASD difference ($p = 0.21$), and reproductive McNemar testing is underpowered ($b = 2, c = 0, p = 0.50$).

B. Selective Prediction

On seed-42 CE logits, validation-set temperature scaling gives $T = 0.875$. The full-test weak-label risk is 0.2053. A confidence threshold of 0.524541, chosen exclusively on 237 validation logits to target approximately 70% coverage, retains 189/263 test crops (coverage 0.7186) at weak-label selective risk 0.1217. Using all 263 ranked test prefixes, the exact AURC is 0.1042 for combined entropy/margin confidence and 0.1036 for max-softmax. The previous AURC 0.126 was a trapezoidal estimate over 14 sparse confidence thresholds, normalized by the observed coverage span (0.285–1.0), rather than all nonempty prefixes; it is superseded by the full-coverage AURC. The historical coverage 0.5399/risk 0.0986 point used a *max-softmax threshold of 0.8*, not the combined-confidence score; Fig. 1 therefore shows two separately labeled curves. The 189/263 retained validation-selected point belongs to the combined-confidence curve and was not fitted to test labels. Neither curve establishes expert-confirmed BBCH correctness.

C. Stage-Gated Retrieval (RQ2)

Table V reports the corrected English-query protocol. Adaptive (P) improves nDCG@5 from 0.343 (B1) to 0.545 and lowers metadata-SIRR@5 from 0.301 to 0.127. Hard filtering minimizes SIRR@5 (0.011) but returns fewer than five documents in 42/420 cases (10%), including two empty lists; missing hits are not counted as incompatible. These numbers

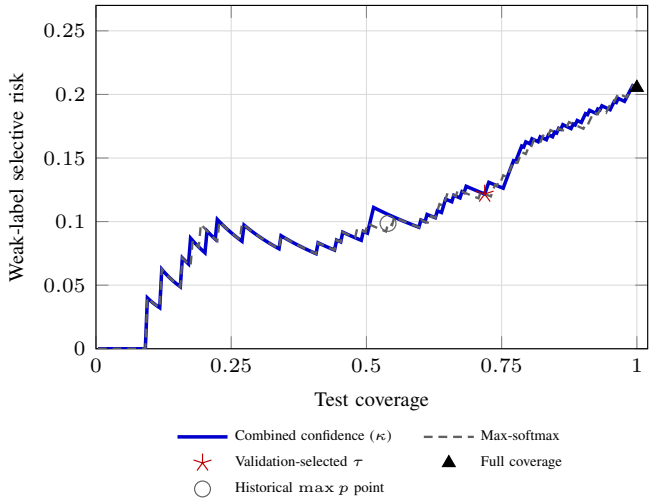


Fig. 1. Full empirical weak-label risk–coverage curves (263 test crops): combined confidence (solid, AURC 0.1042) and max-softmax (dashed, AURC 0.1036). The star marks the validation-selected combined-confidence threshold ($\tau = 0.524541$; 189/263 retained); the hollow circle marks the earlier illustrative max-softmax cutoff (0.8; 142/263 retained). AURC averages risk over all nonempty top-k prefixes; no expert BBCH labels were used.

TABLE V
CORRECTED RETRIEVAL ON 420 METADATA-LABELED SCENARIOS (199 CROPS / 63 PARENT PHOTOS); FIXED B3 IS *untuned*.

Method	P@5	R@5	nDCG@5	SIRR@5↓	Full@5
B0: Ungated	0.277	0.235	0.283	0.218	100%
B1: Query exp.	0.373	0.241	0.343	0.301	100%
B2: Hard filter	0.350	0.340	0.374	**0.011**	90%
B3: Fixed soft	0.409	0.317	0.465	0.175	100%
P: Adaptive	**0.485**	**0.380**	**0.545**	0.127	100%
Reference oracle	0.614	0.453	0.661	0.070	100%

cannot be compared directly with the legacy Vietnamese-query protocol.

On original-parent-photo-clustered paired tests (5,000 samples), P minus B1 reaches ΔnDCG@5 $+0.202$, 95% CI $[+0.165, +0.243]$, Holm-adjusted sign-flip $p = 0.0024$; metadata-SIRR decreases by 0.174, CI $[-0.195, -0.152]$ (Table VI). These estimates treat crops of one original photograph as dependent.

An *independent validation-only tuning step* selects fixed $\beta = 0.35$ from 359 validation scenarios (180 crops, 59 parent photos). Frozen on test, it reaches nDCG@5 0.612 / SIRR@5 0.101; a top-1-belief variant reaches 0.617/0.098, both better than combined-confidence Adaptive (0.545/0.127). The latter is therefore not established as the best routing strategy. On 84 adjacent-error scenarios, P nDCG is 0.435, compared with 0.385 (B1) and 0.444 (untuned B3). On only six nonadjacent errors, P metadata-SIRR is 0.433 versus 0.100 for hard filtering; the subset is too small for general inference.

VI. DISCUSSION

RQ1. The historical three-seed ResNet-18 test accuracy is about 0.800 against six morphology-derived pseudo-stages, exceeding the majority reference (0.335). Since training seeds

TABLE VI
PAIRED DIFFERENCES WITH 95% PARENT-PHOTO BOOTSTRAP CIS AND HOLM-ADJUSTED p-VALUES.

Pair	Metric	Δ (95% CI)	p_{Holm}
P − B1	nDCG@5	$+.202$ $[+.165, +.243]$	.0024
P − B1	SIRR@5	$-.174$ $[-.195, -.152]$	.0024
Fixed − P	nDCG@5	$+.066$ $[+.050, +.083]$	.0012

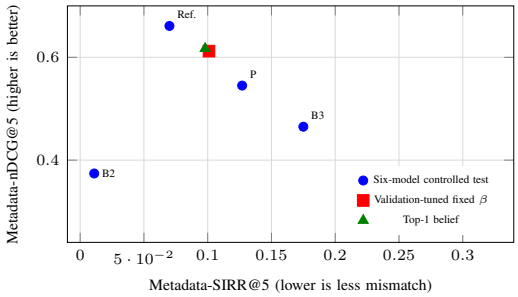


Fig. 2. Ranking versus *metadata-defined stage mismatch*, not independent agronomic safety. The validation-tuned and top-1 variants are shown separately from untuned B3.

share one split and only 24 reproductive and ten grain-filling test crops, repeatability on the fixed subset is not independent field replication. Calibration and selective-risk results are evaluated against these weak labels.

RQ2. In the corrected English-language retrieval setting, adaptive gating improves metadata-defined ranking and mismatch over B1. Hard filtering has the lowest mismatch but sacrifices complete Top-5 coverage. Validation-tuned fixed-soft and top-1 belief outperform the original combined-confidence adaptive approach; the paper therefore evaluates a trade-off, *not* universal superiority of adaptivity.

RQ3. The 420 scenarios are conditional on topic-stage eligibility (199 of 263 test crops) and nested within 63 parent photographs. Retrieval relevance and SIRR are derived from the same stage-compatibility annotations consumed by reranking. Consequently, these are controlled mechanism-consistency scores, not independent recommendation-quality or field-safety measurements.

VII. LIMITATIONS AND ETHICAL CONSIDERATIONS

The labels are deterministic, BBCH-inspired pseudo-stages, not agronomist-confirmed phenological observations. One ResNet-18 backbone and one fixed split cannot establish generalization across fields, seasons, cultivars or acquisition devices. Rare-stage support is limited, and grouping by aHash reduces detected duplication without demonstrating new-field independence. Retrieval's metadata-derived judgments may favor algorithms using that metadata; no independent expert relevance or agronomic-safety reference is available. These are intrinsic evidence boundaries, not uncompleted experiments.

Ethical considerations. Crop photographs may expose location or personal details; publish source images only under the relevant permissions and remove identifying metadata absent

consent. The system is not an autonomous agronomic decision authority: regulated or high-impact recommendations require qualified human judgment.

VIII. Conclusion

CropState-Vision evaluates six weakly supervised rice pseudo-stages and stage-aware retrieval with explicit uncertainty and provenance controls. ResNet-18 accuracy is approximately 0.800 on a fixed split. On 420 scenarios, Adaptive improves metadata nDCG@5 from 0.343 to 0.545 over query expansion and lowers metadata-SIRR@5 from 0.301 to 0.127. However, validation-tuned fixed-soft reaches 0.612/0.101 and top-1 belief reaches 0.617/0.098, exceeding combined-confidence Adaptive on these metadata measures. Hard filtering minimizes mismatch at the cost of underfilled Top-5 outputs. These results support an internal evaluation framework, not a claim of expert-confirmed BBCH accuracy, agronomic usefulness or field safety.

Data and code availability. The 2,900-crop benchmark is derived from RiceSEG [19]; the 173 English chunks are machine-curated from IRRI public knowledge material [20]. Underlying photos and source text retain their original access conditions. Reproduction code and permitted configuration/provenance records are at `github.com/khov4nfs/cropstate`.

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